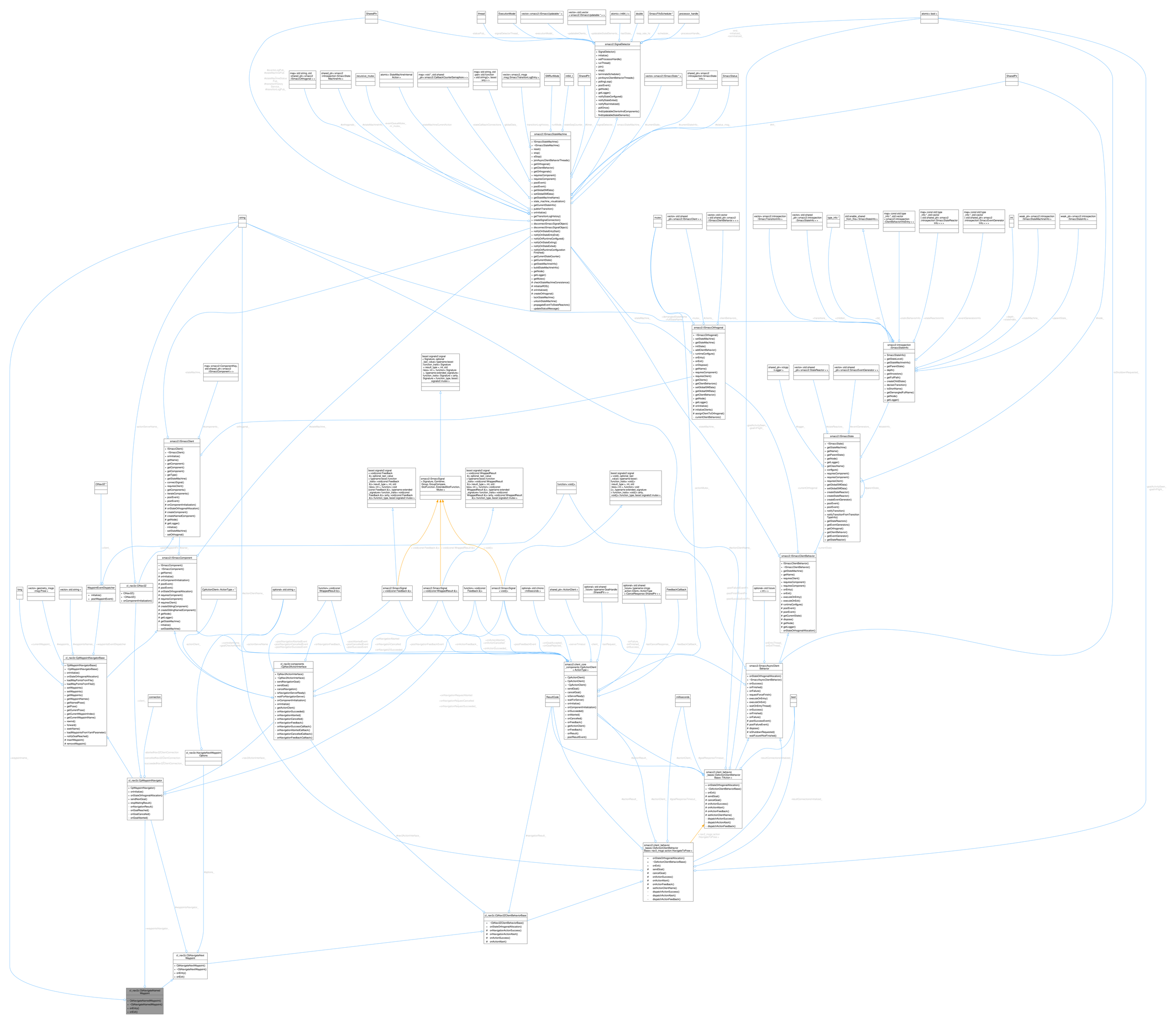
SharedPtr
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< int64_t >
double
SmaccFifoScheduler *
processor_handle
atomic< bool >
statusPub_
signalDetectorThread_
executionModel_
updatableClients_
updatableStateElements_
lastState_
loop_rate_hz
scheduler_
processorHandle_
end_initialized_
wasInitialized_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
+ pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
#eventLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccStateMachineInfo >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2 ::introspection::SmaccStateInfo >
SmaccStatus
SharedPtr
atomic< bool >
#orthogonals_
#stateMachineInfo_
#eventQueueMutex_
cm_mutex_
#stateMachineCurrentAction
#stateCallbackConnections
#globalData_
transitionLogHistory_
#runMode_
#stateSeqCounter_
#timer_
#signalDetector_
#smaccStateMachine_
#currentState_
#currentStateInfo_
#status_msg_
#rm_
#rm_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
mutex
vector< std::shared _ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfo(Entry) > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
int
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection ::SmaccStateInfo >
string
#stateMachine_
->demangledStateName
->fullStateName
->mutex_
->clients_
->clientBehaviors_
->transitions_
->children_
->id_
->staticBehaviorInfo
->stateReactorsInfo
->eventGeneratorsInfo
->depth
->stateIndex_
->stateMachine_
->parentState_
#node_
->isShutdownRequested_
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
#stateMachine_
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
boost::signals2::signal < Signature, optional _last_value, typename boost ::function_traits< Signature >::result_type, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost ::signals2::mutex >
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
->logger_
->stateReactors_
->eventGenerators_
->stateInfo_
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransitions()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
->currentOrthogonal
->parentState_
->actionServerName_
->components_
->orthogonal_
#stateMachine_
->goalActivityTimer_
->goalInFlight_
->actionClientName_
->currentState
GNav22 *
long
vector< geometry_msgs ::msg::Pose >
vector< std::string >
WaypointEventDispatcher
+ initialize()
+ postWaypointEvent()
cl_nav2z::ClNav2Z
+ ClNav2Z()
+ ~ClNav2Z()
+ onComponentInitialization()
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
CpActionClient< ActionType > *
#actionClientName_
optional< std::string >
function< void(const WrappedResult &)>
smacc2::SmaccSignal < void(const Feedback &)>
smacc2::SmaccSignal < void(const WrappedResult &)>
function< void(const Feedback &)>
smacc2::SmaccSignal < void()>
optional< std::chrono ::milliseconds >
shared_ptr< ActionClient >
optional< std::shared _future< typename rclcpp ::action::Client< ActionType >::CancelResponse::SharedPtr > >
optional< std::shared _future< typename goal_handle ::SharedPtr > >
FeedbackCallback
optional< std::future < int > >
->currentWaypoint_
#waypoints_
#waypointsNames_
->waypointsEventDispatcher
cl_nav2z::ClWaypointNavigatorBase
+ ClWaypointNavigatorBase()
+ ~ClWaypointNavigatorBase()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
+ insertWaypoint()
# removeWaypoint()
->actionClient_
->controllerName_
->goalCheckerName_
->actionServerName_
->postNavigationAbortedEvent
->postNavigationCancelledEvent
->postNavigationSucceedEvent
->onNavigationFeedback_
->postAbortedEvent
->postCancelledEvent
->postSuccessEvent
->onNavigationAborted
->onNavigationCancelled
->onNavigationSucceeded
->postNavigationFeedbackEvent
->onActionFeedback_
->onActionAborted_
->onActionCancelled
->onActionSucceeded_
->postFeedbackEvent
->onGoalAccepted_
->onGoalRejected_
->serverTimeout
->client_
->lastRequest_
->onFailure_
->onFinished_
->onSuccess_
->lastCancelResponse_
->feedbackCallback_
->postFailureEvent
->postFinishEvent
->postSuccessEvent
->onEntryThread_
->onExitThread_
cl_nav2z::components _CpNavActionInterface
+ CpNavActionInterface()
+ ~CpNavActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ oninitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
+ onNavigationSuccessCallback()
+ onNavigationAbortedCallback()
+ onNavigationCancelledCallback()
+ onNavigationFeedbackCallback()
smacc2::client_core _components::CpActionClient < ActionType >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ oninitialize()
+ onComponentinitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
# waitFutureIfNotFinished()
->navigationRequestAborted
->navigationRequestCancelled
->navigationRequestSucceeded_
ResultCode
milliseconds
bool
->waypointname_
cl_nav2z::ClWaypointNavigator
+ ClWaypointNavigator()
+ oninitialize()
+ onStateOrthogonalAllocation()
+ sendNextGoal()
+ stopWaitingResult()
- onNavigationResult()
- onGoalReached()
- onGoalCancelled()
- onGoalAborted()
connection
cl_nav2z::ClNavigateNext Waypoint
+ ClNavigateNextWaypoint()
+ ~ClNavigateNextWaypoint()
+ onEntry()
+ onExit()
cl_nav2z::ClNavigateNamed Waypoint
+ ClNavigateNamedWaypoint()
+ ~ClNavigateNamedWaypoint()
+ onEntry()
+ onExit()
->abortedNav2ZClientConnection_
->cancelledNav2ZClientConnection_
->succeededNav2ZClientConnection_
cl_nav2z::NavigateNextWaypoint Options
->nav2ActionInterface_
->options_
->waypointsNavigator_
->waypointsNavigator_
->waypointsNavigator_
->nav2ActionInterface_
->actionResult_
->actionClient_
->goalResponseTimeout_
->navigationResult_
smacc2::client_behavior _bases::CbActionClientBehavior Base< TAction >
+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()
->resultConnectionsInitialized_
->actionResult_
->actionClient_
->goalResponseTimeout_
->navigationResult_
smacc2::client_behavior _bases::CbActionClientBehavior Base< nav2_msgs::action::NavigateToPose >
+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()
->actionMutex_
->nav_msgs::action ::NavigateToPose >
cl_nav2z::ClNav2ZClientBehaviorBase
+ ~ClNav2ZClientBehaviorBase()
+ onStateOrthogonalAllocation()
# onNavigationActionSuccess()
# onNavigationActionAbort()
# onActionSuccess()
# onActionAbort()
boost::signals2::signal < void(const Feedback &), optional_last_value < typename boost::function_traits< void(const Feedback &)>::result_type >, int, std ::less< int >, function< void(const Feedback &)>, typename extended_signature< function_traits< void(const Feedback &)>::arity, void(const Feedback &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(const WrappedResult &), optional_last_value < typename boost::function_traits< void(const WrappedResult &)>::result_type >, int, std ::less< int >, function< void(const WrappedResult &)>, typename extended_signature< function_traits< void(const WrappedResult &)>::arity, void(const WrappedResult &)>::function_type, boost::signals2::mutex >
function< void()>
boost::signals2::signal < void(), optional_last_value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
->navigationResult_
->actionMutex_
->navigationResult_
->actionResult_
->actionClient_
->goalResponseTimeout_
->resultConnectionsInitialized_
->nav2ActionInterface_
->options_
->waypointsNavigator_
->waypointsNavigator_
->nav2ActionInterface_
->actionResult_
->actionClient_
->goalResponseTimeout_
->resultConnectionsInitialized_